

Limits Graphically and Numerically

ANSWER KEY



Estimating limits from a table

- 1 A function g satisfies $g(2.9) = 6.7$, $g(2.99) = 6.97$, $g(2.999) = 6.997$, $g(3.1) = 7.3$, $g(3.01) = 7.03$, $g(3.001) = 7.003$. Estimate $\lim_{x \rightarrow 3} g(x)$.

Both sides approach 7: $\lim_{x \rightarrow 3} g(x) = 7$.

- 2 Using a table of values with x near 0 (radians), estimate $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2}$ to two decimal places (the known exact value is $\frac{1}{2}$).

Values approach 0.50 from both sides, consistent with the exact limit $\frac{1}{2}$.

Reading limits from a graph

- 3 The graph below shows $f(x) = \frac{x^2 - 1}{x - 1}$, which is undefined at $x = 1$. Based on the simplified form $f(x) = x + 1$ ($x \neq 1$), state $\lim_{x \rightarrow 1} f(x)$.

$\lim_{x \rightarrow 1} f(x) = 2$ (even though $f(1)$ itself is undefined — a removable discontinuity).

- 4 A function h has $\lim_{x \rightarrow 2^-} h(x) = 5$ and $\lim_{x \rightarrow 2^+} h(x) = 5$, but $h(2) = 8$. State $\lim_{x \rightarrow 2} h(x)$ and explain whether h is continuous at $x = 2$.

$\lim_{x \rightarrow 2} h(x) = 5$, since both one-sided limits agree. But h is NOT continuous at $x = 2$, since $\lim_{x \rightarrow 2} h(x) = 5 \neq h(2) = 8$ — this is a removable discontinuity.

One-sided limits and limits that fail to exist

- 5 For $f(x) = \frac{x}{|x|}$, find:

a $\lim_{x \rightarrow 0^+} f(x) = 1$

b $\lim_{x \rightarrow 0^-} f(x) = -1$

c $\lim_{x \rightarrow 0} f(x)$ does not exist

- 6 A piecewise function is defined by $f(x) = x^2$ for $x < 1$, and $f(x) = 3 - x$ for $x \geq 1$. Find $\lim_{x \rightarrow 1^-} f(x)$, $\lim_{x \rightarrow 1^+} f(x)$, and state whether $\lim_{x \rightarrow 1} f(x)$ exists.

$\lim_{x \rightarrow 1^-} f(x) = 1^2 = 1$. $\lim_{x \rightarrow 1^+} f(x) = 3 - 1 = 2$. Since the one-sided limits disagree ($1 \neq 2$), $\lim_{x \rightarrow 1} f(x)$ does not exist.

Algebraic evaluation

7 Evaluate each limit by direct substitution or factoring:

a $\lim_{x \rightarrow 4} (3x^2 - 5x + 1) = 3(16) - 20 + 1 = 29$

b $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} = \lim_{x \rightarrow 3} (x + 3) = 6$

c $\lim_{x \rightarrow -2} \frac{x^2 + 5x + 6}{x + 2} = \lim_{x \rightarrow -2} (x + 3) = 1$

8 Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{x+9} - 3}{x}$ by rationalizing the numerator.

Multiply by $\frac{\sqrt{x+9} + 3}{\sqrt{x+9} + 3}$: $\frac{x}{x(\sqrt{x+9} + 3)} = \frac{1}{\sqrt{x+9} + 3} \rightarrow \frac{1}{6}$.

Limits at infinity from a graph

9 The graph shows $f(x) = \frac{2x}{x+1}$, which has a horizontal asymptote. Use the graph to estimate $\lim_{x \rightarrow \infty} f(x)$, then confirm algebraically by dividing numerator and denominator by x .

The graph flattens toward height 2 as $x \rightarrow \infty$. Algebraically: $\frac{2x}{x+1} = \frac{2}{1+1/x} \rightarrow \frac{2}{1+0} = 2$.

Limits involving infinite discontinuities

10 For $f(x) = \frac{1}{x-3}$, find $\lim_{x \rightarrow 3^-} f(x)$ and $\lim_{x \rightarrow 3^+} f(x)$, and state whether $\lim_{x \rightarrow 3} f(x)$ exists.

As $x \rightarrow 3^-$, $x - 3 \rightarrow 0^-$, so $f(x) \rightarrow -\infty$. As $x \rightarrow 3^+$, $x - 3 \rightarrow 0^+$, so $f(x) \rightarrow +\infty$. Since the one-sided limits are not equal (in fact neither is finite), $\lim_{x \rightarrow 3} f(x)$ does not exist.

11 Evaluate $\lim_{x \rightarrow \infty} \frac{3x^2 + 1}{x^2 - 4}$, using the fact that the numerator and denominator have the same degree.

For equal-degree rational functions, the limit is the ratio of leading coefficients: $\lim_{x \rightarrow \infty} \frac{3x^2 + 1}{x^2 - 4} = \frac{3}{1} = 3$.

Estimating limits with a calculator-style table

12 Construct a mental table of values for $f(x) = \frac{e^x - 1}{x}$ at $x = -0.1, -0.01, 0.01, 0.1$ (all close to 0), and use it to estimate $\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$.

The values all cluster near 1 (e.g. $f(0.1) \approx 1.052$, $f(-0.1) \approx 0.952$, both approaching 1 as $x \rightarrow 0$):

$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1$ (a standard limit related to the derivative of e^x at 0).

Limits with piecewise graphical behavior

- 13 A graph shows $f(x) = 2$ for $x < 0$, $f(x) = x + 2$ for $0 \leq x < 3$, and $f(x) = 8 - x$ for $x \geq 3$. Find $\lim_{x \rightarrow 0} f(x)$ and $\lim_{x \rightarrow 3} f(x)$, stating whether each exists.

At $x = 0$: $\lim_{x \rightarrow 0^-} f(x) = 2$, $\lim_{x \rightarrow 0^+} f(x) = 0 + 2 = 2$: limit exists, equals 2. At $x = 3$: $\lim_{x \rightarrow 3^-} f(x) = 3 + 2 = 5$, $\lim_{x \rightarrow 3^+} f(x) = 8 - 3 = 5$: limit exists, equals 5.

- 14 Evaluate $\lim_{x \rightarrow 2} \frac{x^2 - 5x + 6}{x^2 - 4}$ by factoring both the numerator and denominator.

$$\frac{(x-2)(x-3)}{(x-2)(x+2)} = \frac{x-3}{x+2} \rightarrow \frac{2-3}{2+2} = -\frac{1}{4}.$$

Multiple choice — easy

- 15 Find $\lim_{x \rightarrow -3} (x^2 + 3x - 4)$.

- a. -3
- b. -4 ✓
- c. 4
- d. -5

Direct substitution: $1(-3)^2 + 3(-3) - 4 = -4$.

- 16 Find $\lim_{x \rightarrow 4} (5x^2 + 5x + 9)$.

- a. -109
- b. 108
- c. 109 ✓
- d. 110

Direct substitution: $5(4)^2 + 5(4) + 9 = 109$.

17 Find $\lim_{x \rightarrow 3} (4x^2 - x + 8)$.

a. 41 ✓

b. 42

c. 40

d. -41

Direct substitution: $4(3)^2 - (3) + 8 = 41$.

18 Find $\lim_{x \rightarrow 3} (5x^2 + 3x - 5)$.

a. 48

b. 50

c. -49

d. 49 ✓

Direct substitution: $5(3)^2 + 3(3) - 5 = 49$.

19 Find $\lim_{x \rightarrow 3} (5x^2 + 7)$.

a. -52

b. 51

c. 52 ✓

d. 53

Direct substitution: $5(3)^2 + 7 = 52$.

20 Find $\lim_{x \rightarrow 1} (x^2 + 2x - 9)$.

a. -6 ✓

b. -7

c. -5

d. 6

Direct substitution: $1(1)^2 + 2(1) - 9 = -6$.

21 Find $\lim_{x \rightarrow 4} (5x^2 - 8x + 5)$.

a. 54

b. 53 ✓

c. -53

d. 52

Direct substitution: $5(4)^2 - 8(4) + 5 = 53$.

22 Find $\lim_{x \rightarrow -1} (5x^2 - 6x - 8)$.

a. 3 ✓

b. 4

c. -3

d. 2

Direct substitution: $5(-1)^2 - 6(-1) - 8 = 3$.

23 Find $\lim_{x \rightarrow -3} (2x^2 - 9x + 8)$.

a. 52

b. -53

c. 54

d. 53 ✓

Direct substitution: $2(-3)^2 - 9(-3) + 8 = 53$.

24 Find $\lim_{x \rightarrow -2} (x^2 - 4x + 6)$.

- a. 17
- b. 19
- c. -18
- d. 18 ✓

Direct substitution: $1(-2)^2 - 4(-2) + 6 = 18$.

25 Find $\lim_{x \rightarrow 4} (4x^2 + 3x + 2)$.

- a. 77
- b. 78 ✓
- c. 79
- d. -78

Direct substitution: $4(4)^2 + 3(4) + 2 = 78$.

26 Find $\lim_{x \rightarrow 3} (4x^2 + 3x - 5)$.

- a. 39
- b. 40 ✓
- c. 41
- d. -40

Direct substitution: $4(3)^2 + 3(3) - 5 = 40$.

27 Find $\lim_{x \rightarrow -4} (3x^2 - 6x - 4)$.

- a. -68
- b. 69
- c. 67
- d. 68 ✓

Direct substitution: $3(-4)^2 - 6(-4) - 4 = 68$.

28 Find $\lim_{x \rightarrow 1} (2x^2 - x - 3)$.

- a. 2
- b. -3
- c. -2 ✓
- d. -1

Direct substitution: $2(1)^2 - (1) - 3 = -2$.

29 Find $\lim_{x \rightarrow 0} (4x^2 + 9x - 5)$.

- a. -6
- b. -4
- c. -5 ✓
- d. 5

Direct substitution: $4(0)^2 + 9(0) - 5 = -5$.

Multiple choice — medium

30 Evaluate $\lim_{x \rightarrow 0} \frac{(x)(x-1)}{x}$ by factoring and cancelling.

- a. -3
- b. 1
- c. -1 ✓
- d. 3

Cancel the common factor: $\frac{(x)(x-1)}{x} = x - 1$ for $x \neq 0$, so the limit is $0 - (1) = -1$.

31 Evaluate $\lim_{x \rightarrow -2} \frac{(x+2)(x-5)}{x+2}$ by factoring and cancelling.

- a. 7
- b. -7 ✓
- c. -5
- d. -9

Cancel the common factor: $\frac{(x+2)(x-5)}{x+2} = x - 5$ for $x \neq -2$, so the limit is $-2 - (5) = -7$.

32 Evaluate $\lim_{x \rightarrow 2} \frac{(x-2)(x-5)}{x-2}$ by factoring and cancelling.

- a. 3
- b. -1
- c. -5
- d. -3 ✓

Cancel the common factor: $\frac{(x-2)(x-5)}{x-2} = x - 5$ for $x \neq 2$, so the limit is $2 - (5) = -3$.

33 Evaluate $\lim_{x \rightarrow 5} \frac{(x-5)(x-6)}{x-5}$ by factoring and cancelling.

- a. -1 ✓
- b. 1
- c. 11
- d. -3

Cancel the common factor: $\frac{(x-5)(x-6)}{x-5} = x - 6$ for $x \neq 5$, so the limit is $5 - (6) = -1$.

34 Evaluate $\lim_{x \rightarrow -4} \frac{(x+4)(x+3)}{x+4}$ by factoring and cancelling.

- a. -3
- b. -1 ✓
- c. -7
- d. 1

Cancel the common factor: $\frac{(x+4)(x+3)}{x+4} = x + 3$ for $x \neq -4$, so the limit is $-4 - (-3) = -1$.

35 Evaluate $\lim_{x \rightarrow -4} \frac{(x+4)(x-5)}{x+4}$ by factoring and cancelling.

a. -11

b. -7

c. -9 ✓

d. 9

Cancel the common factor: $\frac{(x+4)(x-5)}{x+4} = x - 5$ for $x \neq -4$, so the limit is $-4 - (5) = -9$.

36 Evaluate $\lim_{x \rightarrow -4} \frac{(x+4)(x+5)}{x+4}$ by factoring and cancelling.

a. 3

b. 1 ✓

c. -9

d. -1

Cancel the common factor: $\frac{(x+4)(x+5)}{x+4} = x + 5$ for $x \neq -4$, so the limit is $-4 - (-5) = 1$.

37 Evaluate $\lim_{x \rightarrow -1} \frac{(x+1)(x-6)}{x+1}$ by factoring and cancelling.

a. -9

b. -5

c. 7

d. -7 ✓

Cancel the common factor: $\frac{(x+1)(x-6)}{x+1} = x - 6$ for $x \neq -1$, so the limit is $-1 - (6) = -7$.

38 Evaluate $\lim_{x \rightarrow 6} \frac{(x-6)(x+4)}{x-6}$ by factoring and cancelling.

a. -10

b. 8

c. 10 ✓

d. 12

Cancel the common factor: $\frac{(x-6)(x+4)}{x-6} = x+4$ for $x \neq 6$, so the limit is $6 - (-4) = 10$.

39 Evaluate $\lim_{x \rightarrow -5} \frac{(x+5)(x-3)}{x+5}$ by factoring and cancelling.

a. -8 ✓

b. -6

c. 8

d. -10

Cancel the common factor: $\frac{(x+5)(x-3)}{x+5} = x-3$ for $x \neq -5$, so the limit is $-5 - (3) = -8$.

40 Evaluate $\lim_{x \rightarrow 6} \frac{(x-6)(x-3)}{x-6}$ by factoring and cancelling.

a. 1

b. -3

c. 3 ✓

d. 5

Cancel the common factor: $\frac{(x-6)(x-3)}{x-6} = x-3$ for $x \neq 6$, so the limit is $6 - (3) = 3$.

41 Evaluate $\lim_{x \rightarrow 6} \frac{(x-6)(x+3)}{x-6}$ by factoring and cancelling.

a. 11

b. 9 ✓

c. 7

d. -9

Cancel the common factor: $\frac{(x-6)(x+3)}{x-6} = x+3$ for $x \neq 6$, so the limit is $6 - (-3) = 9$.

42 Evaluate $\lim_{x \rightarrow 3} \frac{(x-3)(x+3)}{x-3}$ by factoring and cancelling.

- a. 4
- b. -6
- c. 8
- d. 6 ✓

Cancel the common factor: $\frac{(x-3)(x+3)}{x-3} = x+3$ for $x \neq 3$, so the limit is $3 - (-3) = 6$.

43 Evaluate $\lim_{x \rightarrow 3} \frac{(x-3)(x+1)}{x-3}$ by factoring and cancelling.

- a. -4
- b. 4 ✓
- c. 6
- d. 2

Cancel the common factor: $\frac{(x-3)(x+1)}{x-3} = x+1$ for $x \neq 3$, so the limit is $3 - (-1) = 4$.

44 Evaluate $\lim_{x \rightarrow -5} \frac{(x+5)(x+1)}{x+5}$ by factoring and cancelling.

- a. -4 ✓
- b. -6
- c. -2
- d. 4

Cancel the common factor: $\frac{(x+5)(x+1)}{x+5} = x+1$ for $x \neq -5$, so the limit is $-5 - (-1) = -4$.

Multiple choice — hard

45 For $f(x) = \frac{-1}{x+5}$, find $\lim_{x \rightarrow -5^-} f(x)$.

- a. -5
- b. $+\infty$ ✓
- c. 0
- d. $-\infty$

As $x \rightarrow -5^-$, the denominator $\rightarrow 0^-$, so $f(x) \rightarrow +\infty$.

46 Evaluate $\lim_{x \rightarrow \infty} \frac{6x^2 + 3x}{8x^2 + 8x}$.

- a. $\frac{4}{3}$
- b. $\frac{3}{4}$ ✓
- c. $\frac{7}{8}$
- d. $\frac{2}{3}$

Numerator and denominator have equal degree (2), so the limit is the ratio of leading coefficients: $\frac{6}{8} = \frac{3}{4}$.

47 Evaluate $\lim_{x \rightarrow \infty} \frac{4x^2 + 4x}{2x^2 - 6x}$.

- a. 2 ✓
- b. $\frac{1}{2}$
- c. $\frac{5}{2}$
- d. $\frac{4}{3}$

Numerator and denominator have equal degree (2), so the limit is the ratio of leading coefficients: $\frac{4}{2} = 2$.

48 Evaluate $\lim_{x \rightarrow \infty} \frac{7x^3 + 2x}{6x^2 + 1}$.

a. 0

b. 1

c. does not exist ($+\infty$) ✓

d. $\frac{7}{6}$

The numerator's degree (3) exceeds the denominator's (2), so the quotient grows without bound: the limit does not exist (diverges to $+\infty$).

49 Evaluate $\lim_{x \rightarrow \infty} \frac{5x^2 + 6x}{9x^2 - 3x}$.

a. $\frac{5}{9}$ ✓

b. $\frac{1}{2}$

c. $\frac{9}{5}$

d. $\frac{2}{3}$

Numerator and denominator have equal degree (2), so the limit is the ratio of leading coefficients: $\frac{5}{9} = \frac{5}{9}$.

50 Evaluate $\lim_{x \rightarrow \infty} \frac{7x^2 - 9x}{2x^2 - 6x}$.

a. $\frac{2}{7}$

b. 4

c. $\frac{7}{2}$ ✓

d. $\frac{7}{3}$

Numerator and denominator have equal degree (2), so the limit is the ratio of leading coefficients: $\frac{7}{2} = \frac{7}{2}$.

51 Evaluate $\lim_{x \rightarrow \infty} \frac{6x+3}{7x^2-1}$.

a. does not exist

b. 0 ✓

c. $\frac{6}{7}$

d. 1

The denominator's degree (2) exceeds the numerator's (1), so the quotient shrinks to 0 as $x \rightarrow \infty$.

52 For $f(x) = \frac{1}{x+5}$, find $\lim_{x \rightarrow -5^-} f(x)$.

a. 0

b. $+\infty$

c. -5

d. $-\infty$ ✓

As $x \rightarrow -5^-$, the denominator $\rightarrow 0^-$, so $f(x) \rightarrow -\infty$.

53 For $f(x) = \frac{-1}{x-2}$, find $\lim_{x \rightarrow 2^-} f(x)$.

a. 2

b. 0

c. $-\infty$

d. $+\infty$ ✓

As $x \rightarrow 2^-$, the denominator $\rightarrow 0^-$, so $f(x) \rightarrow +\infty$.

54 Evaluate $\lim_{x \rightarrow \infty} \frac{8x^2-7x}{4x^2+6x}$.

a. 2 ✓

b. $\frac{9}{4}$

c. $\frac{1}{2}$

d. $\frac{8}{5}$

Numerator and denominator have equal degree (2), so the limit is the ratio of leading coefficients: $\frac{8}{4} = 2$.

55 For $f(x) = \frac{-1}{x+6}$, find $\lim_{x \rightarrow -6^-} f(x)$.

- a. -6
- b. 0
- c. $+\infty$ ✓
- d. $-\infty$

As $x \rightarrow -6^-$, the denominator $\rightarrow 0^-$, so $f(x) \rightarrow +\infty$.

56 Evaluate $\lim_{x \rightarrow \infty} \frac{8x+3}{x^2-1}$.

- a. does not exist
- b. 8
- c. 0 ✓
- d. 1

The denominator's degree (2) exceeds the numerator's (1), so the quotient shrinks to 0 as $x \rightarrow \infty$.

57 Evaluate $\lim_{x \rightarrow \infty} \frac{9x^3+2x}{2x^2+1}$.

- a. does not exist ($+\infty$) ✓
- b. 1
- c. $\frac{9}{2}$
- d. 0

The numerator's degree (3) exceeds the denominator's (2), so the quotient grows without bound: the limit does not exist (diverges to $+\infty$).

58 Evaluate $\lim_{x \rightarrow \infty} \frac{8x^3 + 2x}{2x^2 + 1}$.

a. 0

b. 1

c. does not exist ($+\infty$) ✓

d. 4

The numerator's degree (3) exceeds the denominator's (2), so the quotient grows without bound: the limit does not exist (diverges to $+\infty$).

59 Evaluate $\lim_{x \rightarrow \infty} \frac{x^3 + 2x}{5x^2 + 1}$.

a. does not exist ($+\infty$) ✓

b. 1

c. $\frac{1}{5}$

d. 0

The numerator's degree (3) exceeds the denominator's (2), so the quotient grows without bound: the limit does not exist (diverges to $+\infty$).